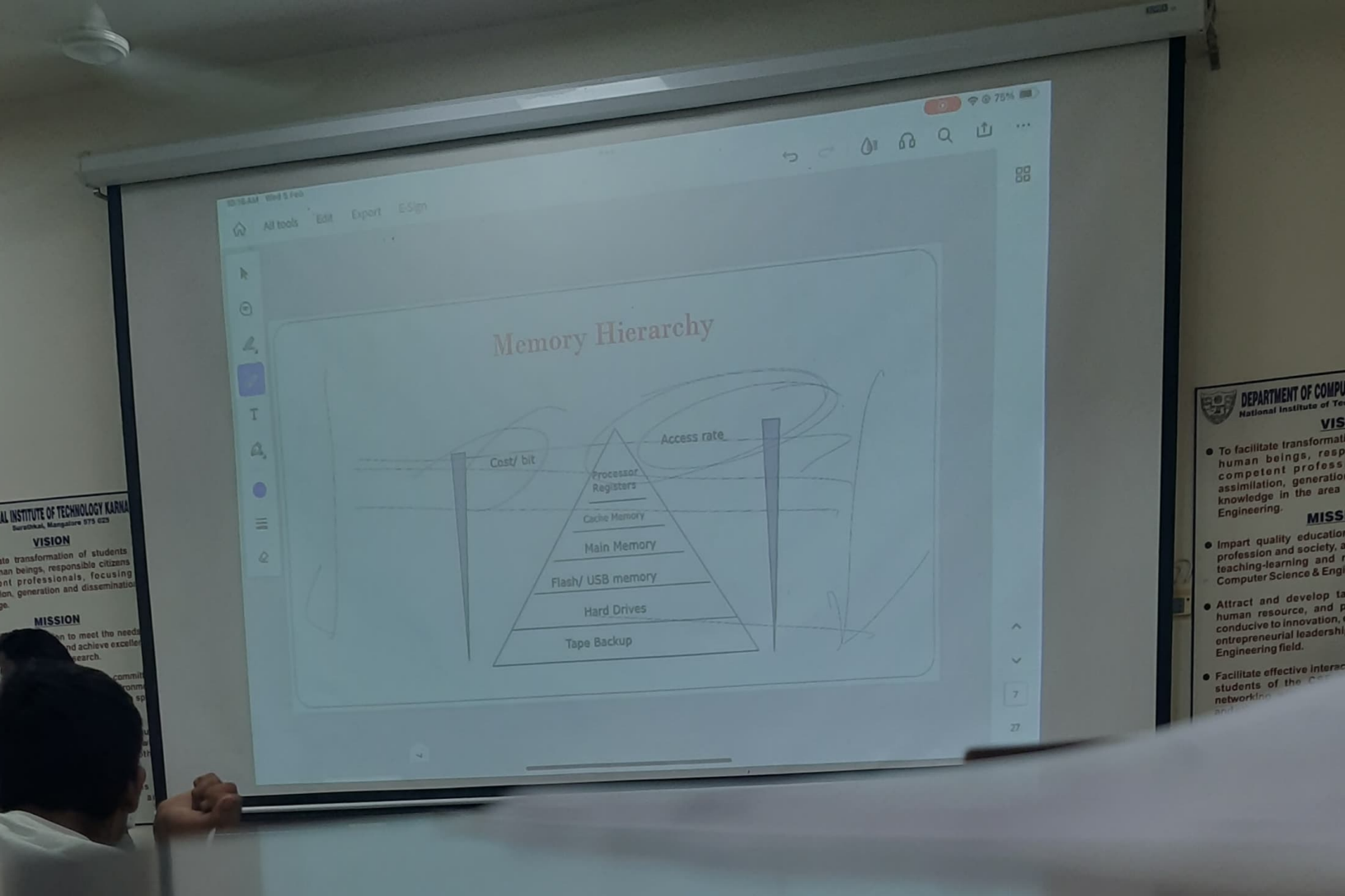

All tools
Edit
Export
E-Sign
Memory Hierarchy
Cost/ bit
Access rate
Processor Registers
Cache Memory
Main Memory
Flash/ USB memory
Hard Drives
Tape Backup
VISION
MISSION

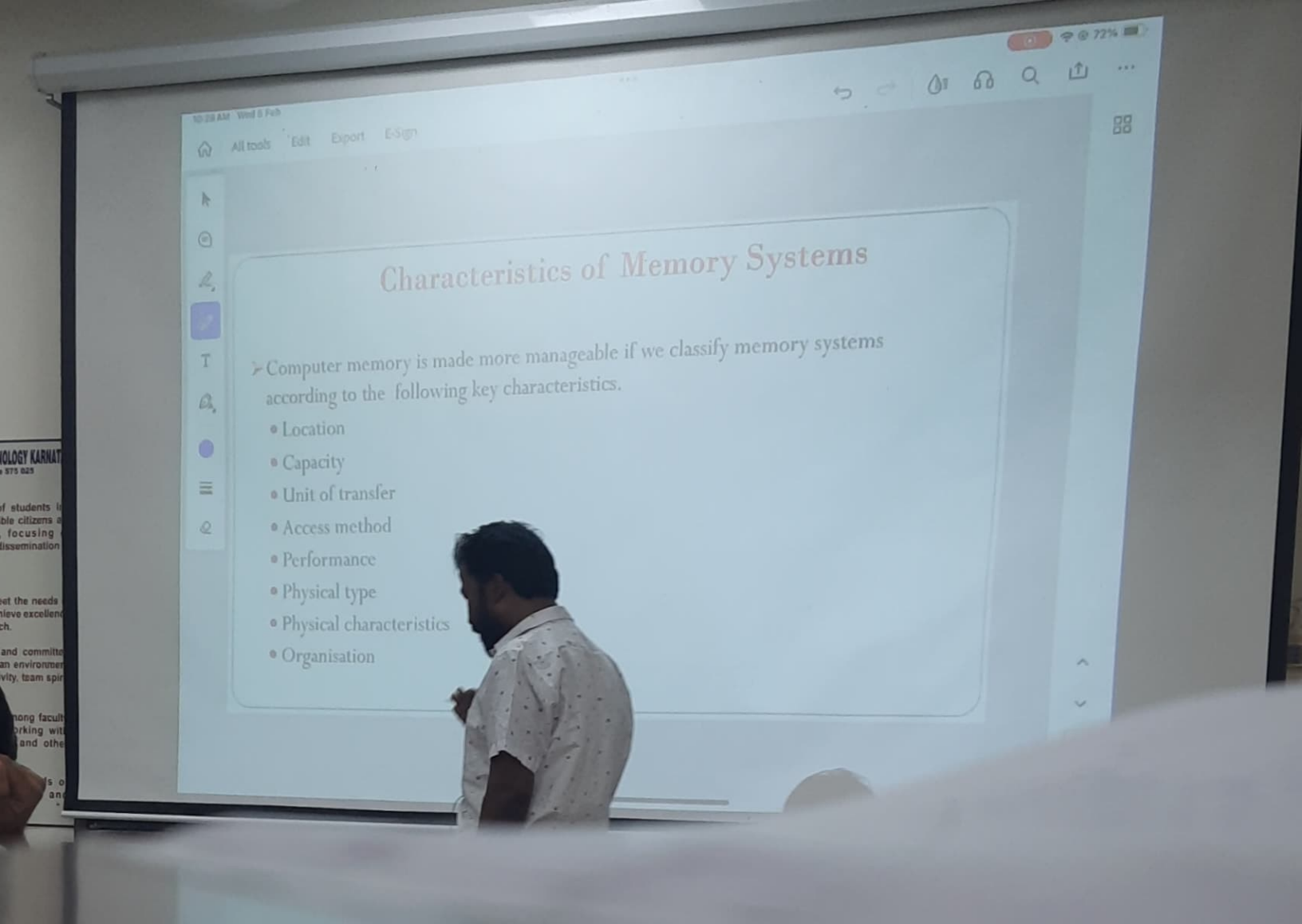
3-Level Memory Classification

A. **Inbound storage / memory**
   - Registers, Cache, Main memory.
   - The class of Internal memory

B. **Outbound Storage**
   - Magnetic Disk
   - CD ROM
   - CD RW
   - DVD + RW
   - DVD + RAM
   - The class of External memory

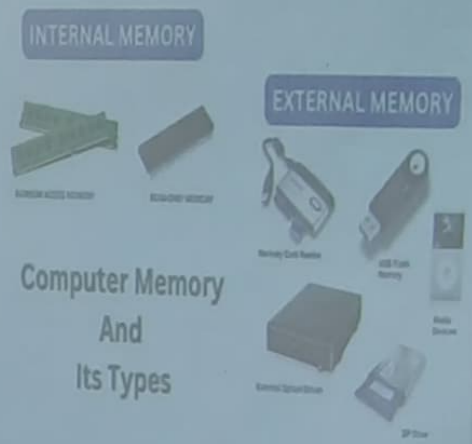
C. **Off-line Storage**
   - Magnetic Tape
   - Hard Disk Drive
   - The class of Backing store

Factors Driving Memory Design


All tools
Edit
Export
E-Sign
Characteristics of Memory Systems
Computer memory is made more manageable if we classify memory systems according to the following key characteristics.
Location
Capacity
Unit of transfer
Access method
Performance
Physical type
Physical characteristics
Organisation

Memory Location

- Location refers to whether memory is **internal** (e.g., processor registers, main memory, cache) and **external** (e.g. optical disks, magnetic disks, tapes) to the computer. Thus, computer locations are two types.
  - ***Internal Memory*** often equated with main memory.
  - ***External Memory*** consists of peripheral storage devices accessible to the processor via I/O controllers.
- CPU
  - ✓ Assumed as another form of internal memory.
  - ✓ The processor requires its own local memory, in the form of registers.



Memory Capacity

- An obvious characteristic of memory is its **capacity**.
- *Internal memory capacity is typically expressed in terms of bytes* (1 byte = 8 bits) or words.
- *External memory capacity is typically expressed in terms of bytes.*
- Word: The natural unit of organisation of memory.
- ***Word Size or Length:*** The size of the word is typically equal to the number of bits used to represent an integer and to the instruction length.
  - Common word lengths are 8, 16, and 32 bits.
  - Unfortunately, there are many exceptions.
    - The CRAY C90 (an older model CRAY supercomputer) has a 64-bit word length but uses a 46-bit integer representation.
    - The Intel x86 architecture has a wide variety of instruction lengths, expressed as multiples of bytes, and a word size of 32 bits.

Memory Capacity Conversion Chart

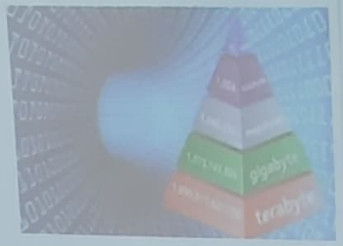
Term (Abbreviation)	Approximate Size
Byte (B)	8 bits
Kilobyte (KB)	1024 bytes / 10^3 bytes
Megabyte (MB)	1024 KB / 10^6 bytes
Gigabyte (GB)	1024 MB / 10^9 bytes
Terabyte (TB)	1024 GB / 10^{12} bytes
Petabyte (PB)	1024 TB / 10^{15} bytes
Exabyte (EB)	1024 PB / 10^{18} bytes
Zettabyte (ZB)	1024 EB / 10^{21} bytes
Yottabyte (YB)	1024 ZB / 10^{24} bytes

Bits & Bytes

- Smallest piece of memory = 1 bit (off/on)
  - 8 bits = 1 byte
  - 4 bytes = 1 word (on 32-bit machines)
  - 8 bytes = 1 word (on 64-bit machines)
- 1 word = number of bits used to store single-precision floating-point number.

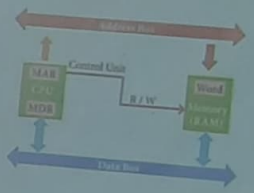
WOMAN LOGIC

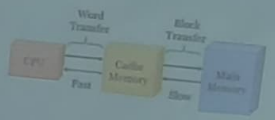
MAN LOGIC



Unit of Transfer

- For main memory, *Unit of transfer is the number of bits read out of or written into memory at a time.*
- The unit of transfer need not equal a word or an addressable unit.
- For Internal Memory: *Usually governed by data bus width*
- For External Memory: *Usually a block which is much larger than a word.*
- Addressable unit
  - Smallest location which can be uniquely addressed.
  - Addressing can be either at Word or at byte level by a system.
  - The relationship between the length in bits A of an address and the number N of addressable units is $2^A = N$.



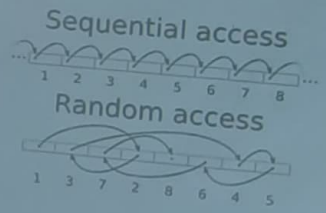


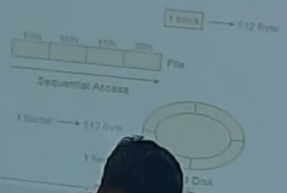
Addressable Unit

- *An addressable unit is a unit of memory in a computer that can be accessed by an address.*
- **Byte:** The smallest addressable unit of memory in most modern computers. A byte is made up of eight bits, each representing a binary value of 0 or 1. This allows a byte to represent 256 unique combinations.
- **Word:** A word is made up of 16 bits.
- **Long word:** A long word is made up of 32 bits.
- *Byte-addressable memory* is a fundamental part of computer architecture and design.
- It allows for more efficient memory use and greater flexibility in handling data.
- *The "**minimum addressable unit**" is the size of a single element of memory - usually a byte, or 8 bits.*
- *Few machines with 12 bits, 16 bits, 32 bits, 36 bits, 64 bits as the smallest unit.*
- The size of the address bus determines the maximum number of bytes that can be addressed.
- For example, a 1-bit address bus can address two locations, a 2-bit address bus can address four locations, and a 20-bit address bus can address 1 MB

Access Methods

- The method of accessing units of data is four types:
  - a) Sequential access,
  - b) Direct access,
  - c) Random access,
  - d) Associative access.





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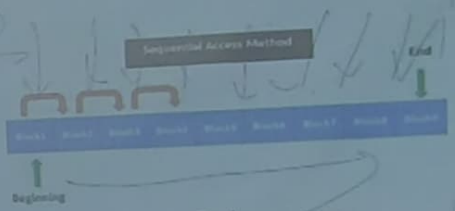
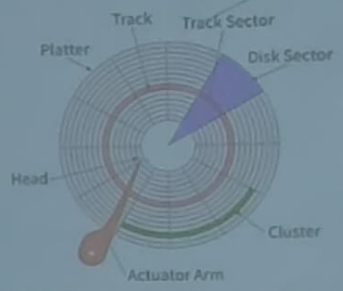
Access Methods

- **Sequential Access**
  - Memory is organized into units of data, called records.
  - Access must be made in a specific linear sequence.
  - Start at the beginning and read through in order.
  - Access time depends on location of data and previous location. Thus, the time to access an arbitrary record is highly variable.
  - **Tape units are sequential access.**



- **Direct Access**
  - direct access involves a shared read–write mechanism.
  - Individual blocks or records have unique address based on physical location.
  - Access is by jumping to vicinity plus sequential search
  - Access time depends on location and previous location.
  - **Disk units are direct access.**



Access Methods

- Random Access
  - Individual addresses identify locations exactly.
  - Access time is independent of location or previous access.
  - Thus, any location can be selected at random and directly addressed and accessed.
  - **Main memory and some cache systems are random access.**

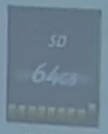
- Associative Access
  - This is a random access type of memory that enables one to make a comparison of desired bit locations within a word for a specified match, and to do this for all words simultaneously.
  - Data is located by a comparison with contents of a portion of the store.
  - Access time is independent of location or previous access.
  - **Cache memories may employ associative access.**

Performance

- Like the capacity, another most important characteristic of memory is *performance*.
- Three performance parameters are used.
  a) Access time (latency)
  b) Memory Cycle time
  c) Transfer Rate
- Access time
  - *Time between presenting the address and getting the valid data.*
  - For random-access memory, this is the time it takes to perform a read or write operation.
  - For non-random-access memory, access time is the time it takes to position the read–write mechanism at the desired location.

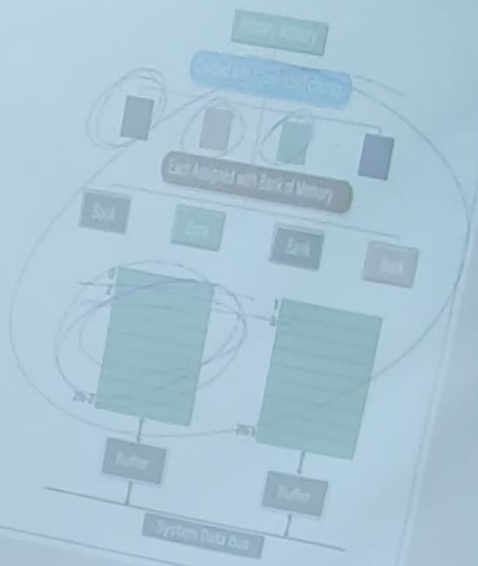
Physical Types

- **Physical type** means a variety of memory.
- The most common today's memory types are
  - a) Semiconductor: RAM
  - b) Magnetic: Disk & Tape
  - c) Optical: CD & DVD
  - d) Magneto-optical: Bubble, Hologram



Physical Characteristics

- Following physical characteristics of data storage are important.
  a) Decay
  b) Volatility
  c) Erasable
  d) Power consumption
- In a volatile memory, information decays naturally or is lost when electrical power is switched off.
- In a nonvolatile memory, information once recorded remains without deterioration until deliberately changed; no electrical power is needed to retain information.
- Magnetic-surface memories are nonvolatile.
- Semiconductor memory may be either volatile or nonvolatile.
- Nonerasable memory cannot be altered, except by destroying the storage unit. Semiconductor memory of this type is known as *read-only memory* (ROM).
- A practical nonerasable memory must also be nonvolatile.

Organisation

- *Physical arrangement of bits into words is meant to an organization.*
- For random-access memory, the **organization** is a key design issue.
- The obvious arrangement is not always used, e.g., interleaved.
- *Interleaved memory is a memory management technique that divides the physical memory into equal-sized chunks.*
- Each chunk is assigned to a particular memory bank and then interleaved.